Nathan Mars | Data Analyst Portfolio Project

Retail Sales Analysis: Pricing, Promotions, and Revenue Optimization

PostgreSQL + Tableau analysis of retail sales performance across time, category, and discount strategy

Objective

This analysis was designed to answer two questions:

1. What patterns drive retail sales performance?
2. How does promotion status and discount bracket affect units sold and sales revenue?

Dataset and Scope

Dataset Overview

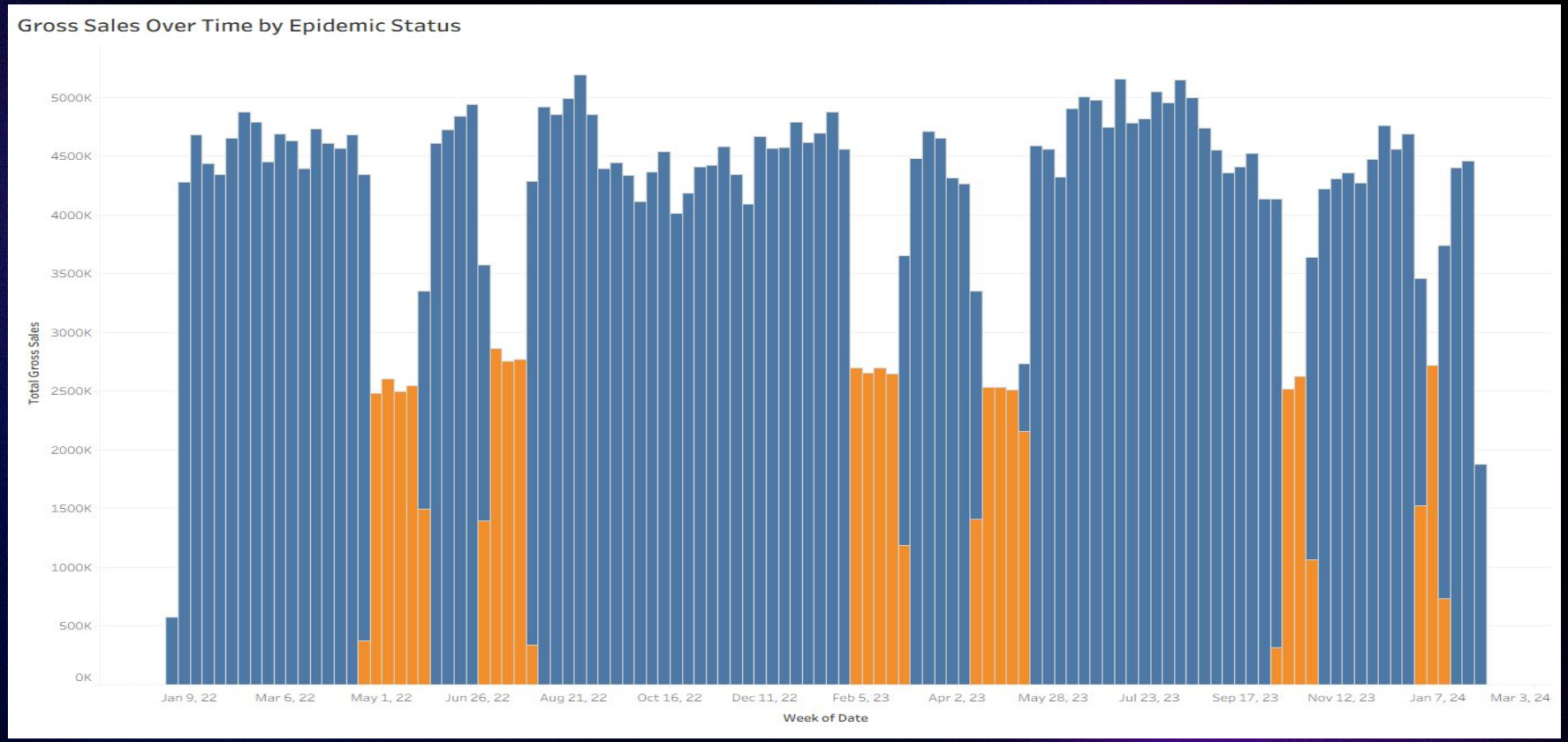
The dataset contains daily retail sales records across multiple stores and products, including:

- Units sold
- Price and Discount Rate
- Promotion Status
- Item Category
- Seasonality and weather condition
- Epidemic Indicator

Scope of Analysis

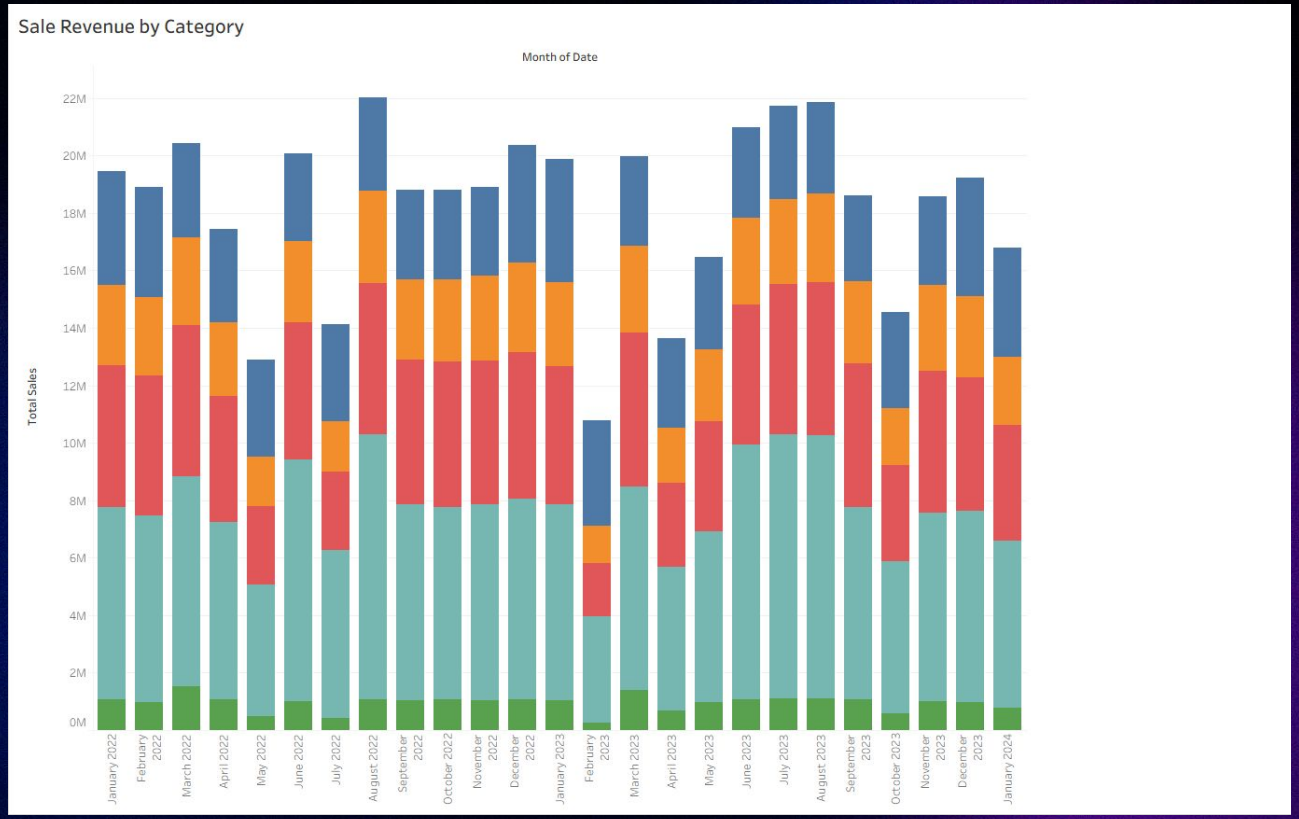
- Overall business context to identify generic trends in sales
- Analysis of discounts and promotional sales
- Promotion analysis was restricted to non-epidemic periods only
  - Time frame and impact of epidemic periods was deemed to not fit historical trends
  - Use of epidemic data would skew expected business trends by nature

Overall Business Context



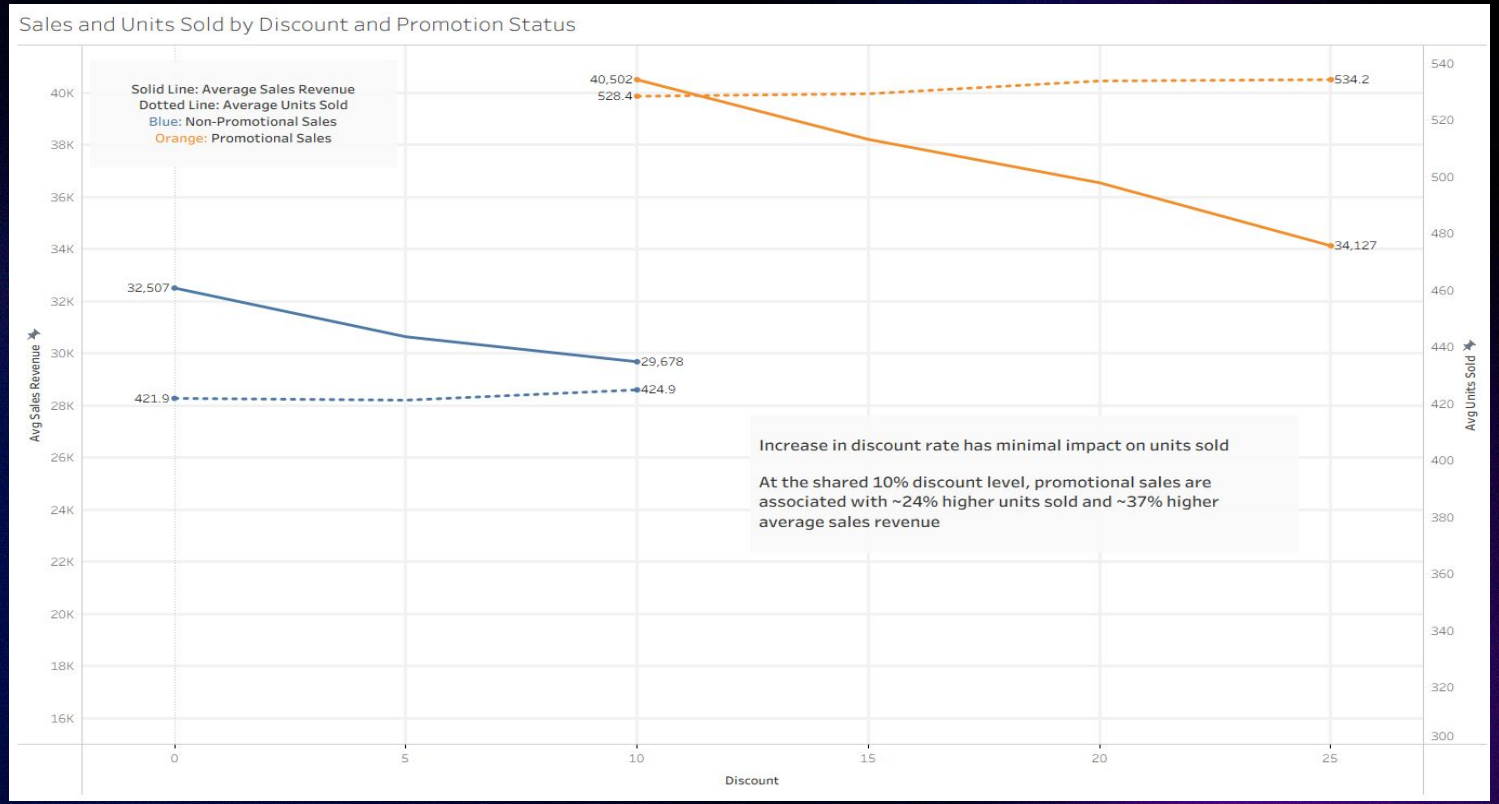
Epidemic-tagged periods naturally result in lower sales, but the overall duration and start/stop periods are not logical based on historical trends. For this reason, epidemic-flagged data is not used during analysis

Overall Business Context- Continued



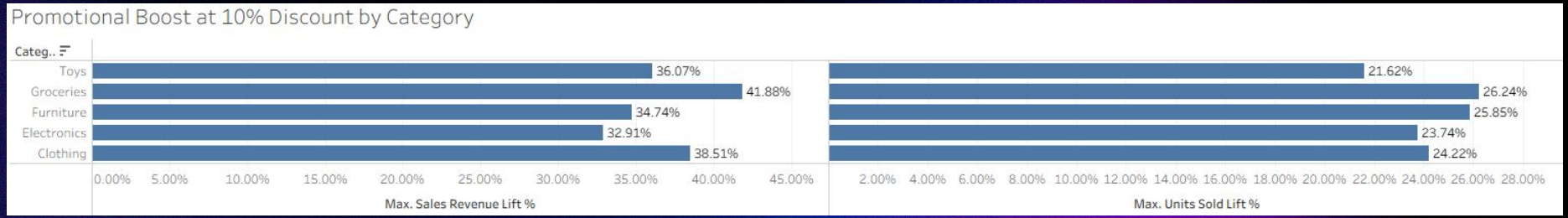
Category sales composition remains relatively stable over time. Epidemic-related dips do not materially change the overall category mix.

Promotion and Discount Analysis



Deep discounts (>15%) did not produce enough volume to offset lower average sales revenue. At the shared 10% discount level, promotional sales were more effective.

Category Validation



To test whether the promotion effect was isolated to a specific category of products, promotion lift at the 10% shared discount level was reviewed by category.

Findings

- All categories show notable lift with promotional status at 10% discount.
- All categories largely fit within our generic trend (8 out of 10 within +/- 10%).

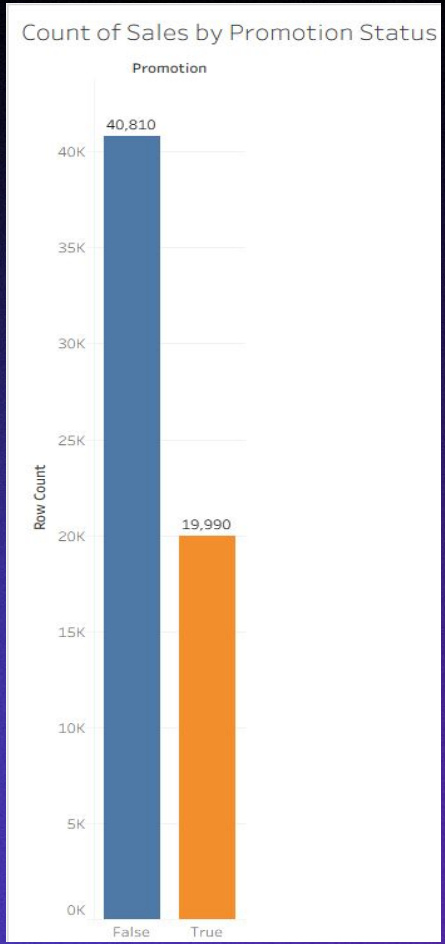
Interpretation

The promotion effect appears broad rather than category-specific, supporting a common pricing recommendation across categories in this dataset.

Scenario Analysis

Reduce Exposure to 25% Discounts

- The observed data shows deeper discounts have negligible impact on units sold, resulting only in reduces sale revenue.
- Promotional sales make up 33% of all total sales.
- Sales within each promotional status follow a uniform distribution.
    - **10,15,20,25% promotional discount each encompass 25% of promotional sales**
- By removing exposure to 25% discounts, the store will realize 0.4% increased revenue at minimum (minimum 5% return on 8% of all sales).



Recommendation

Shift all 25% discounts into 10% promotional discounts

Current Avg Revenue 25 Promo	6,720
Projected Avg Revenue 10 Promo	8,105
Row Count 25 Promo	4,983
Current Total Revenue 25 Promo	33,485,672
Projected Total Revenue As 10 Pr..	40,385,680
Net Increased Revenue	6,900,008
Total Revenue All Non Epidemic	398,665,864
Pct Increased Revenue Of All Sales	1.73

- Shifting all 25% promotional discounts to 10% results in 1.73% increased revenue, or $6.9 million.
- This maintains existing 2:1 non-promotional to promotional ratio to reduce risk of “promotion fatigue.”
- Deeper discounts in general should be avoided as they provide no tangible increase in units sold

Limitations and Next Steps

Limitations

- Recommendation assumes that 2:1 sale status is predominant factor in preventing “promotion fatigue”
  - **Further analysis is needed to determine if distribution across discount brackets has any affect.**
- Promotional status is observational
  - **No insight into why a promotion was active**
- No insight into product placement
  - **Promotional items could be solely placed on aisle end caps which boosts sale rate**

Next Steps

- Specialized discount strategies on a per category basis
  - **Grocery pricing may warrant further analysis due to its stronger performance on promotional status**
- Analyzing discount strategy on a per-season basis
  - **Sales are largely consistent except for Spring, which is ~30% lower.**
  - **Further analysis could yield season specific strategies that further boost revenue**
- Examination of store-level differences to determine store-specific strategies
- Implementation of product cost to evaluate profit (additional data necessary).

Thank you!

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